

CSCI 135 Assignment – Hello Python! Hello World!

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August 23rd/25th, 2023

Due Date: Sunday, 8/27/2023 11:59PM

Points: 20

Topic: Getting Started with Python

The purpose of this assignment is to get you started programming in Python. You will learn how to write, edit, and run your own programs/scripts. You will use a text editor (Notepad++) to write and edit your scripts and you will run your scripts using the (Anaconda) Python Shell.

Submission:

After following the instructions below, please submit the two Python scripts in Moodle. Upload these as separate files – do not .zip them. Name them exactly as follows:

- HelloWorld.py
- GreetingsFromMe.py

Grading:

Each program/script is worth 10 points each for a total of 20 points.

For each program, you will be graded according to the following criteria:

	HelloWorld.py	GreetingsFromMe.py
Program Runs	2	2
Header Comment	2	2
Program Runs Correctly	4	4
Correctly Formatted Output	2	2
Total	10	10

Part 1: Hello World!

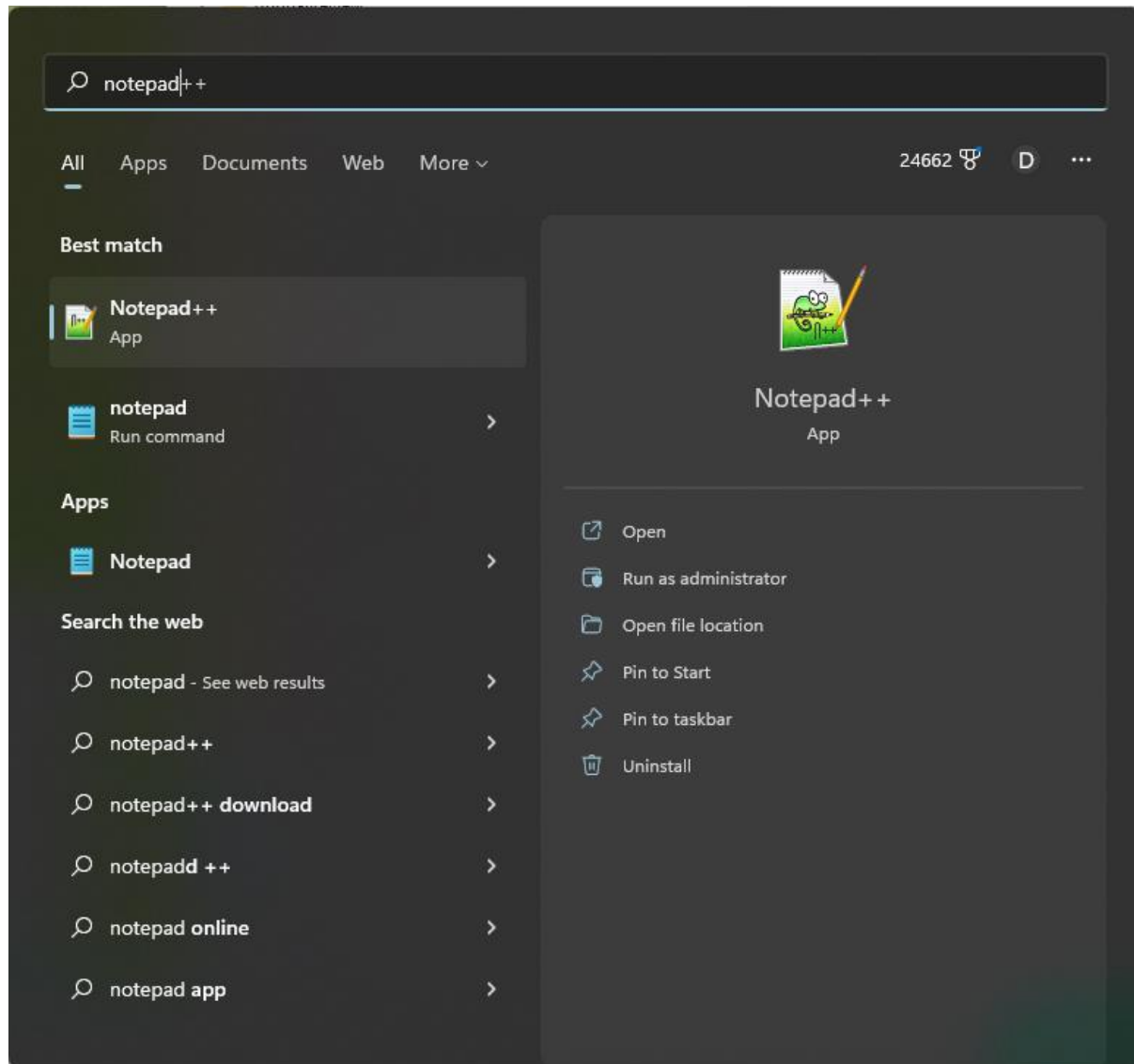
The traditional first program when learning almost any programming language is “Hello World”. (See http://en.wikipedia.org/wiki/Hello_world_program).

Since this may be your first time using the Notepad++ editor and the Anaconda Python shell, I will show you step-by-step what to do on the following pages.

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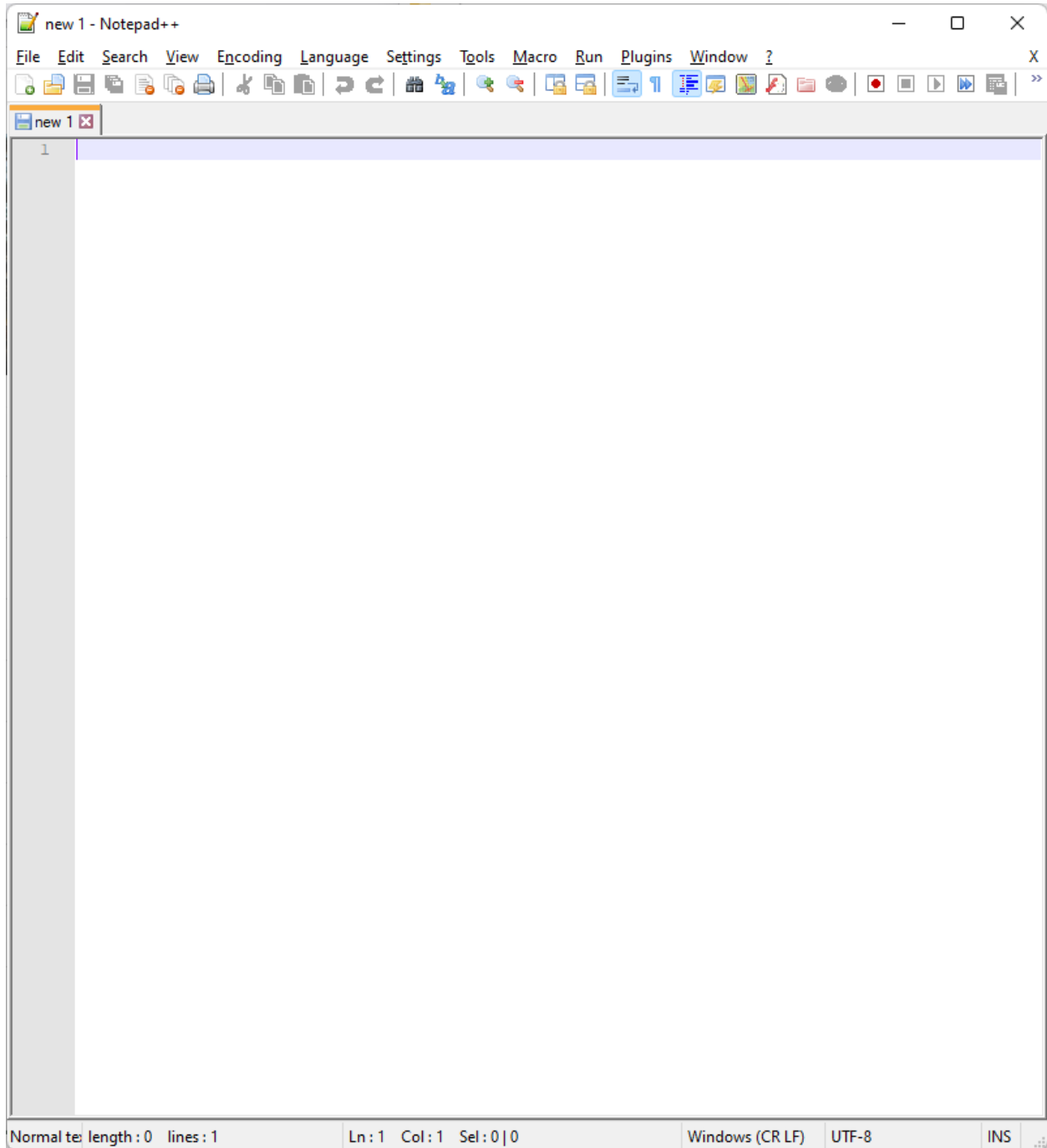
Open the Notepad++ text editor.

You can do this via the Start Menu or by searching. There are other editors you can use. Notepad++ is the one I will use in class, and I recommend that you use it. NOTE: A text editor is not the same as a Word Processor. I do not show the steps for downloading and installing Notepad++. It should be installed in the labs. To install on your personal computer, go here for the download: <https://notepad-plus-plus.org/downloads/> It is only available for Windows computers. Let me know if you need to find an alternative for Mac or other.



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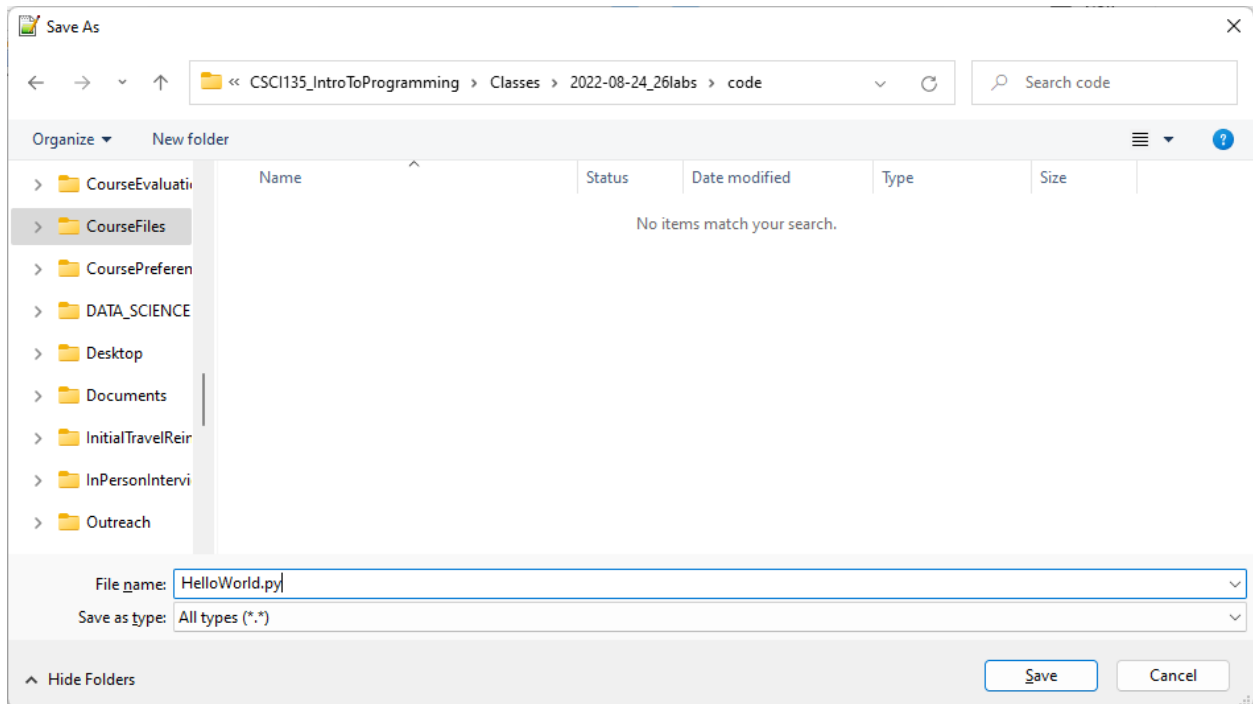
Upon opening Notepad++ you should see that it provides an empty “document” for you:



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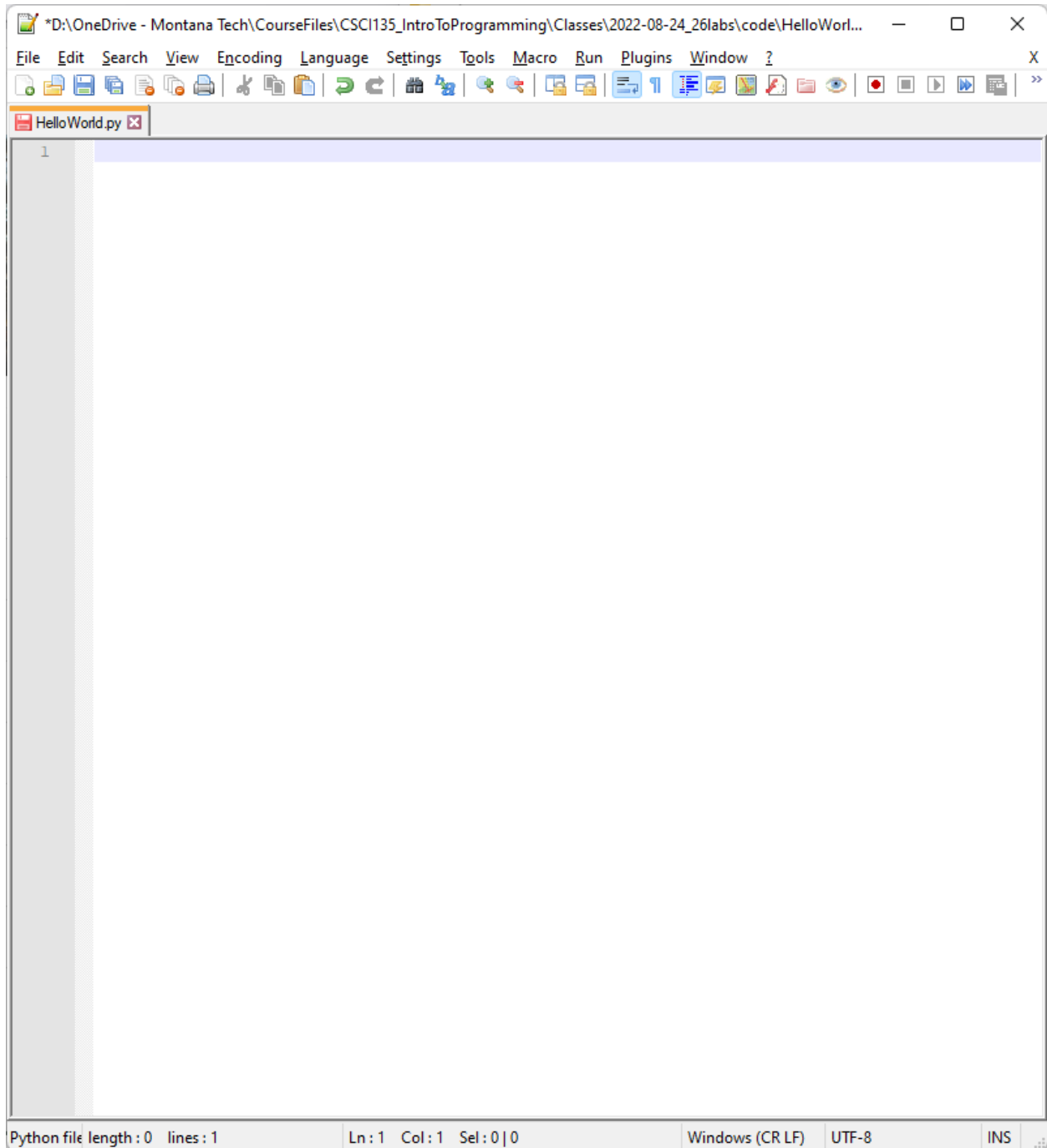
While you could start typing content/code, it is a good idea to first save to a file. Navigate to the location (directory) where you want to save the file, enter the name for it (HelloWorld.py), and press the Save button.

I save my files to my OneDrive class folder so I can access them from campus, home, etc. To do this, I have to configure the OneDrive application on each computer I am working on. You could also save your files to a thumb drive or other “portable” location and carry it with you. You should not save your files to just the local machine in a lab because you might not have access to that machine the next time you need to work on it. If you do save to the local machine while working on a file, make sure you copy the file to a portable or “cloud” location before you leave so you can access it elsewhere.



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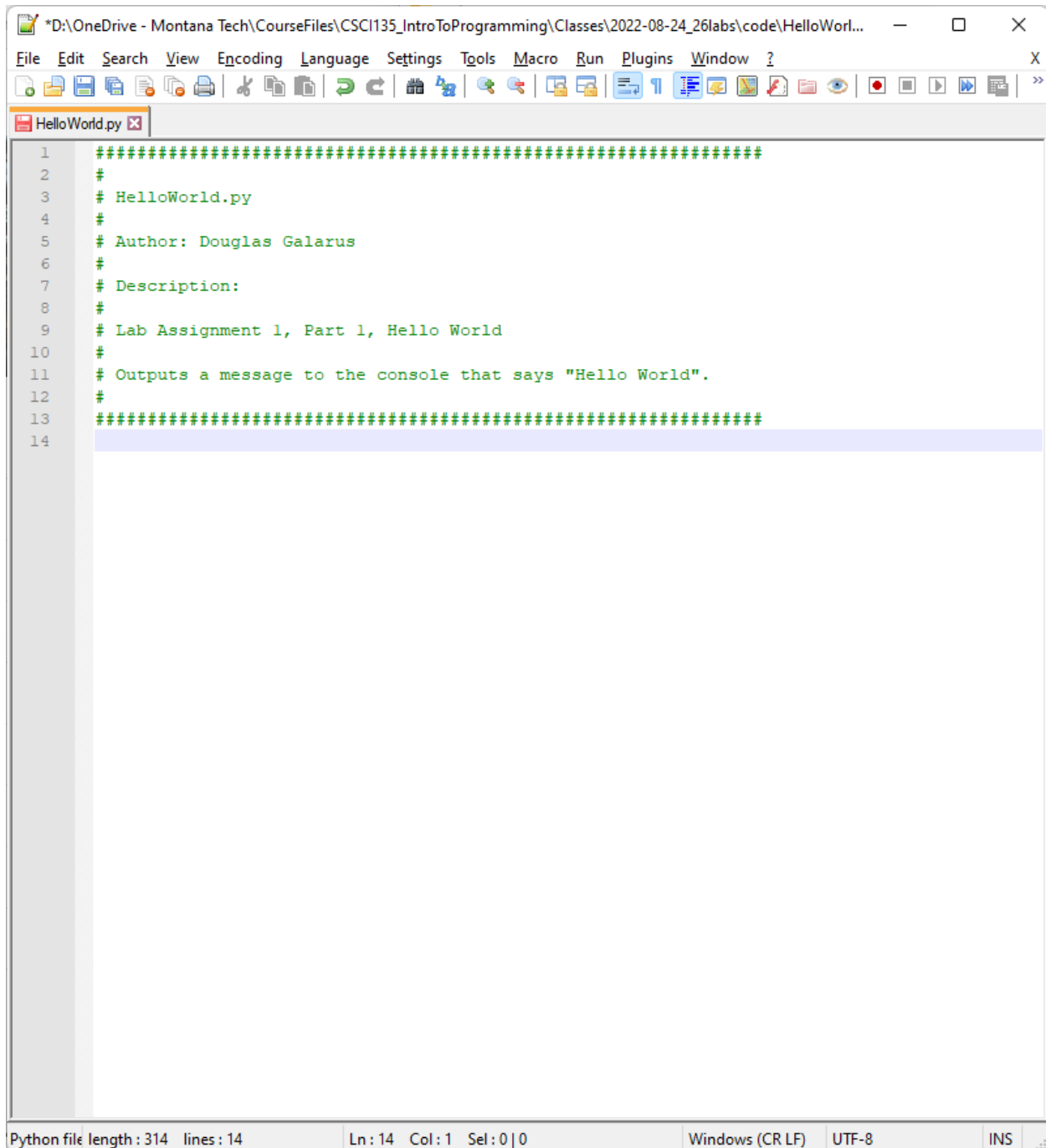
Once named and saved, you should see the path to the file in the title bar of Notepad++. Notice that the path to my file is very long. You should also see indications of whether changes have been made since the last time you saved.



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The first content you should put into your file is a header that indicates the name, author, and description of the file. This “documentation” is helpful in that it provides human-readable/written information about the file that you or someone else can read later to know what the file does and who wrote it. The header is entered using Python “comments”. The comments here are designated with the # symbol at the beginning. This tells Python to ignore what follows on a line that starts with #.

Follow my example below and type a similar header using your name in your HelloWorld.py file for this assignment.



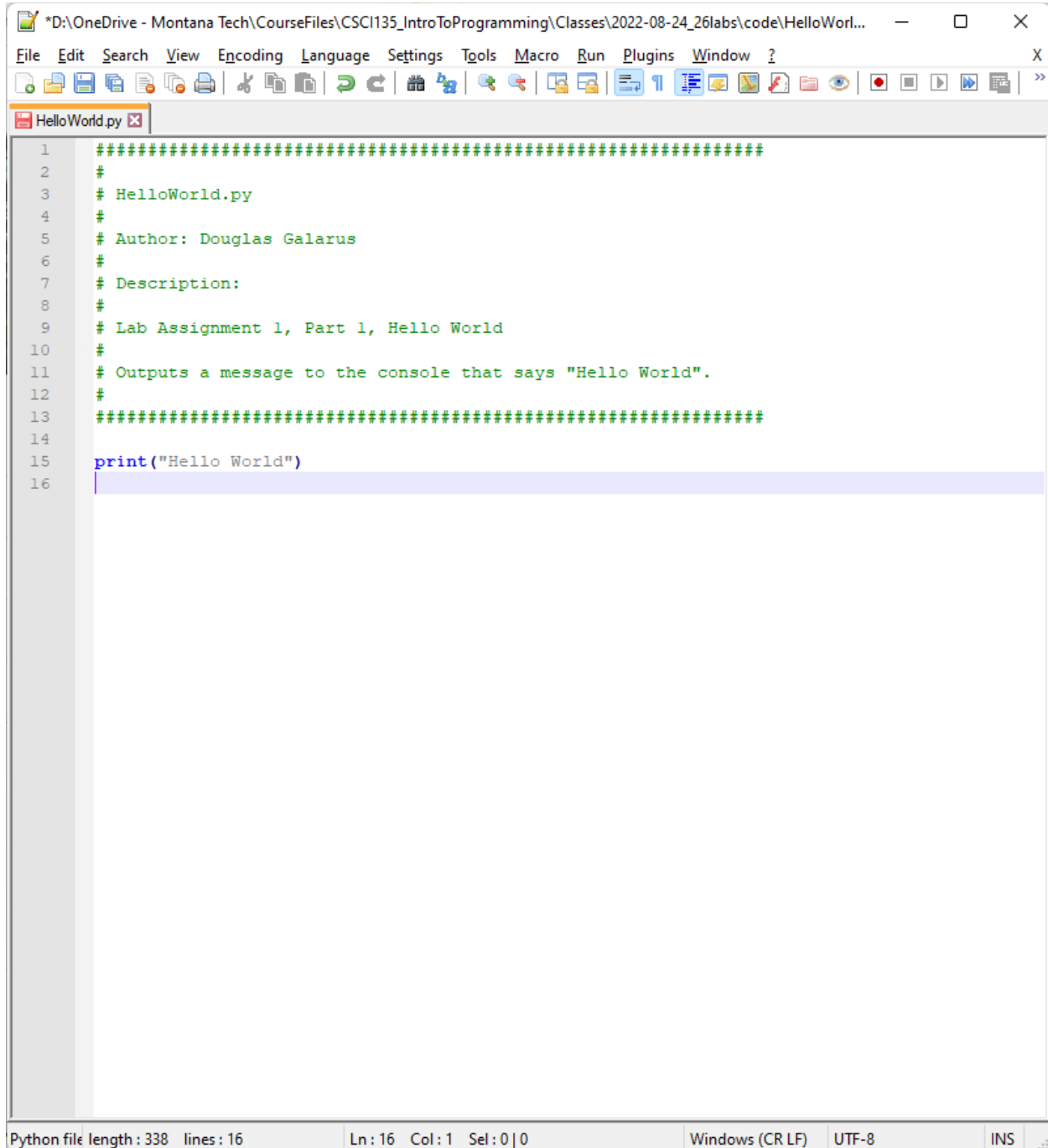
The screenshot shows a Python IDE window titled "HelloWorld.py". The code is as follows:

```
1 #####
2 #
3 # HelloWorld.py
4 #
5 # Author: Douglas Galarus
6 #
7 # Description:
8 #
9 # Lab Assignment 1, Part 1, Hello World
10 #
11 # Outputs a message to the console that says "Hello World".
12 #
13 #####
14
```

The status bar at the bottom indicates: Python file length : 314 lines : 14, Ln: 14 Col: 1 Sel: 0 | 0, Windows (CR LF), UTF-8, INS.

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Enter the “code” that prints “Hello World”. This is a one-liner. You pass the text string “Hello World” to the print function to accomplish this. Notice that Notepad++ colors this line different than the header lines. It recognizes that the header lines are comments and colors them green. It recognizes that print is a function and colors it blue. The string parameter passed to the function is colored grey.



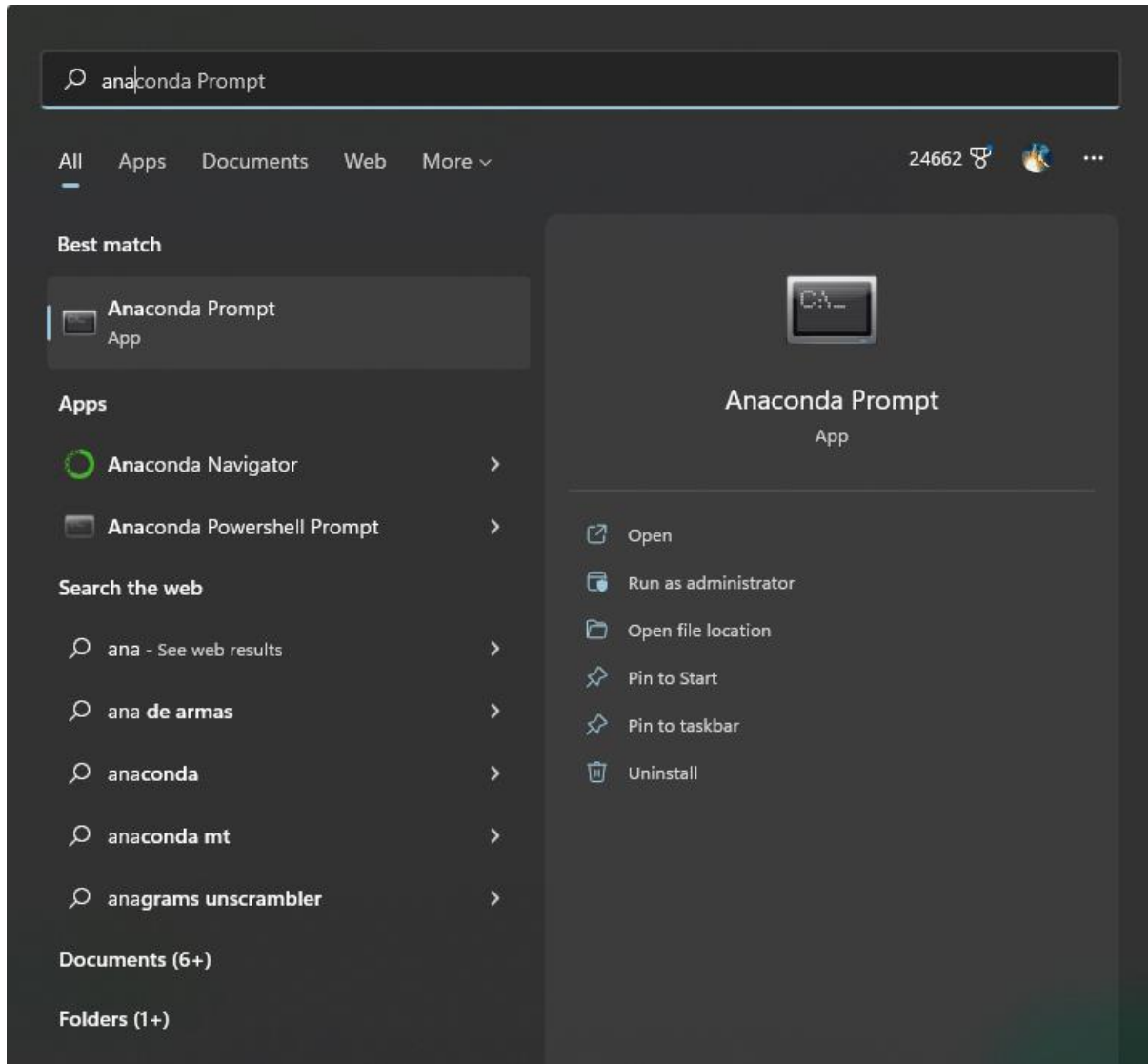
```
1 #####
2 #
3 # HelloWorld.py
4 #
5 # Author: Douglas Galarus
6 #
7 # Description:
8 #
9 # Lab Assignment 1, Part 1, Hello World
10 #
11 # Outputs a message to the console that says "Hello World".
12 #
13 #####
14
15 print("Hello World")
16
```

Python file length : 338 lines : 16 Ln : 16 Col : 1 Sel : 0 | 0 Windows (CR LF) UTF-8 INS

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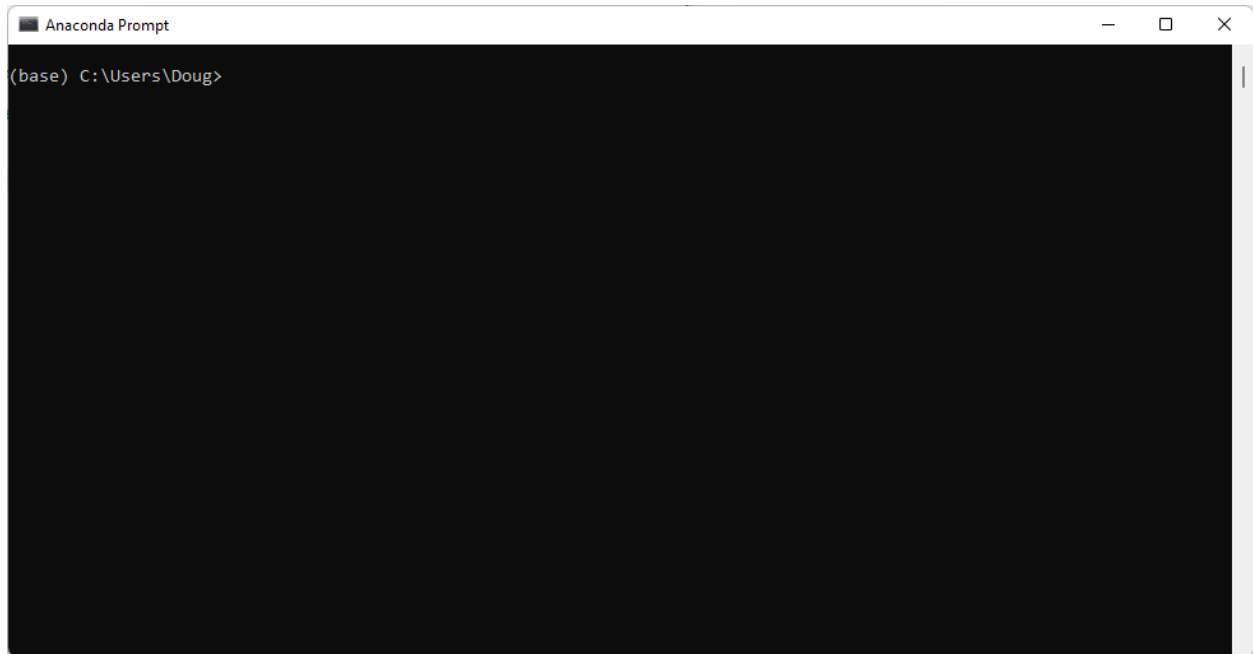
To run your Python script, you will need to open the Anaconda Prompt. You can get to this through the start menu or by searching, assuming it is installed on the machine you are working on. The Anaconda distribution should be installed on our lab computers. To install it on your personal computer, follow the instructions that I provide separately.

Open the Anaconda Prompt.



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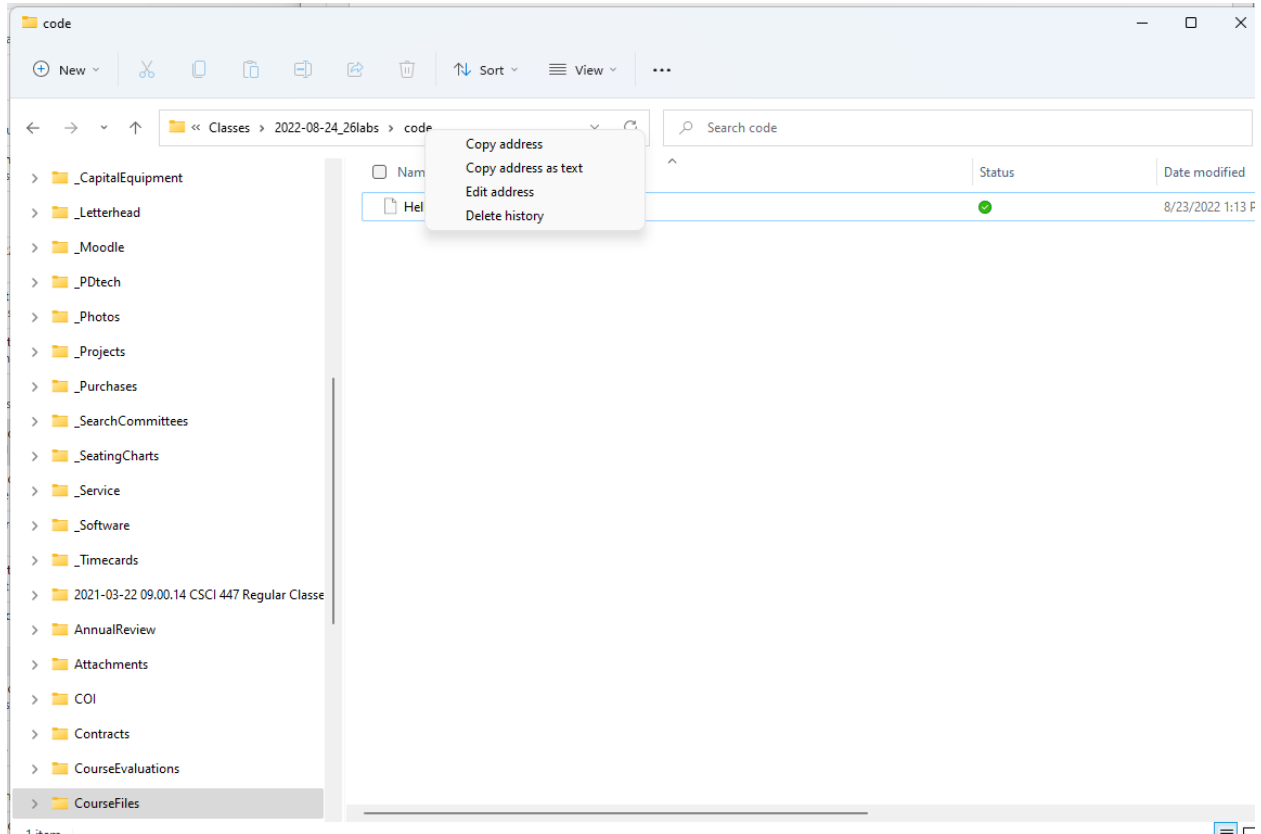
The Anaconda prompt will show you a “command line prompt” indicating what directory it is in. This is important as you need to know what directory your Python script is in to run it, and it probably won’t be in the directory that opens by default.



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You might not have memorized the exact path to the location where you saved your file, but you should know how to find via the File Explorer.

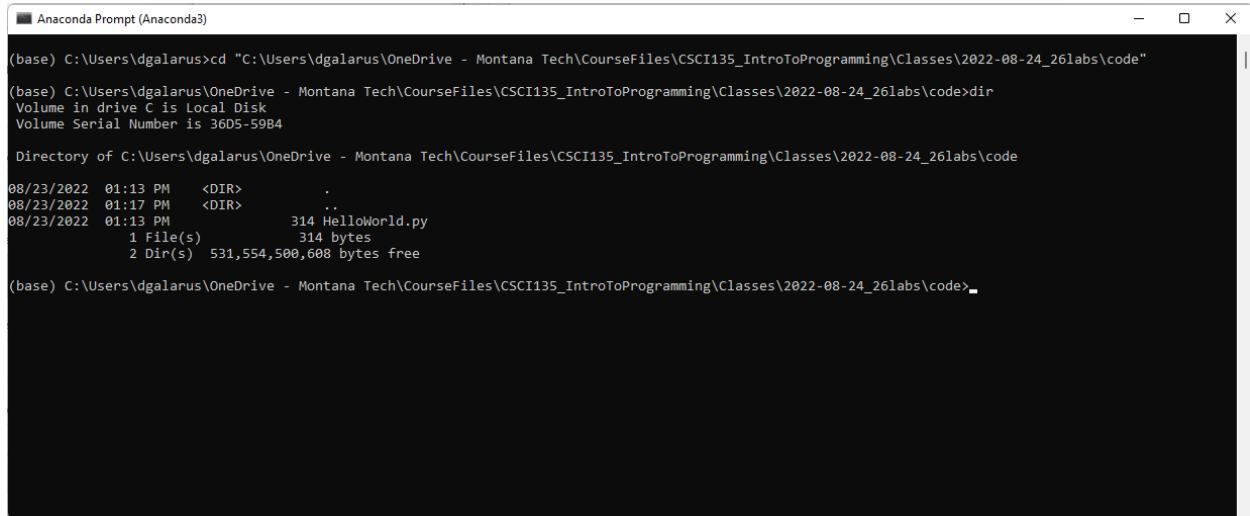
Open the File Explorer, navigate to the folder containing your Python script, and copy the address from the navigation bar. Copy it as text.



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Within the Anaconda prompt, navigate to the directory containing your Python script. Use the `cd` command followed by the path to the directory. Put the path in quotes, which helps to handle spaces and other special characters in a long path. You can paste the path if you copied it in the prior step. After issuing the `cd` command, notice that the command prompt changes to reflect that path.

Once in the directory, you should see your Python script listed when using the `dir` command.

A screenshot of an Anaconda Prompt window. The title bar reads "Anaconda Prompt (Anaconda3)". The terminal shows the following commands and output:

```
(base) C:\Users\dgalarus>cd "C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code"
(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>dir
Volume in drive C is Local Disk
Volume Serial Number is 36D5-59B4

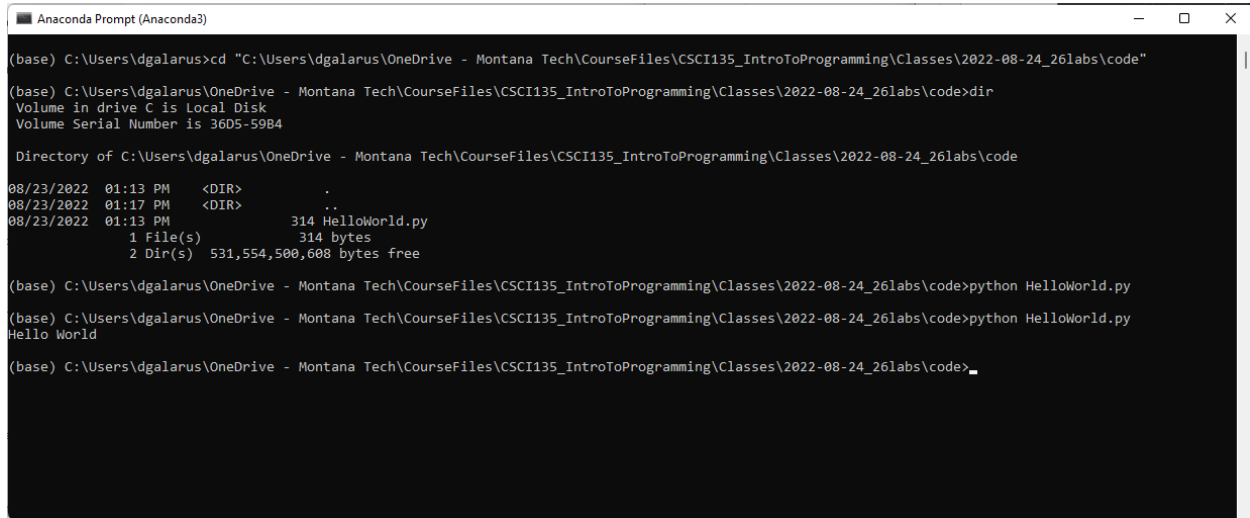
Directory of C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code

08/23/2022  01:13 PM    <DIR>          .
08/23/2022  01:17 PM    <DIR>          ..
08/23/2022  01:13 PM                314 HelloWorld.py
               1 File(s)                314 bytes
               2 Dir(s)  531,554,500,608 bytes free

(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>_
```

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Now you can run your script. Issue the command “python HelloWorld.py”. This should execute your Python script and display the Hello World message. Note that in the following screenshot I tried running the script before saving the file with the print command. Nothing was displayed. After saving, the correct output was shown. Check that the correct output is shown. If not, find the source of your problem and fix it.



```
Anaconda Prompt (Anaconda3)

(base) C:\Users\dgalarus>cd "C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code"

(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>dir
Volume in drive C is Local Disk
Volume Serial Number is 36D5-59B4

Directory of C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code

08/23/2022  01:13 PM    <DIR>          .
08/23/2022  01:17 PM    <DIR>          ..
08/23/2022  01:13 PM                314 HelloWorld.py
               1 File(s)                314 bytes
               2 Dir(s)  531,554,500,608 bytes free

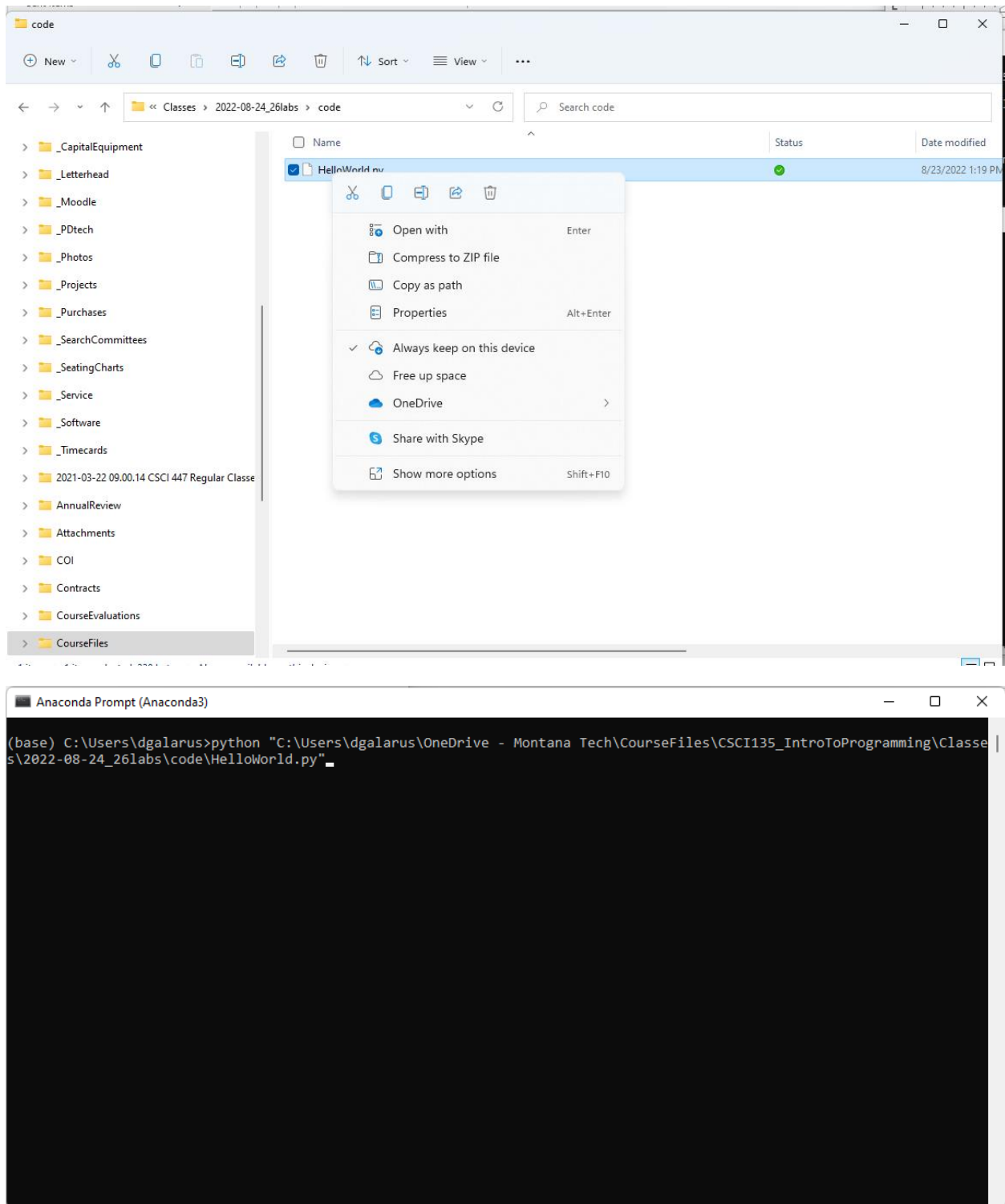
(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>python HelloWorld.py

(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>python HelloWorld.py
Hello World

(base) C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code>_
```

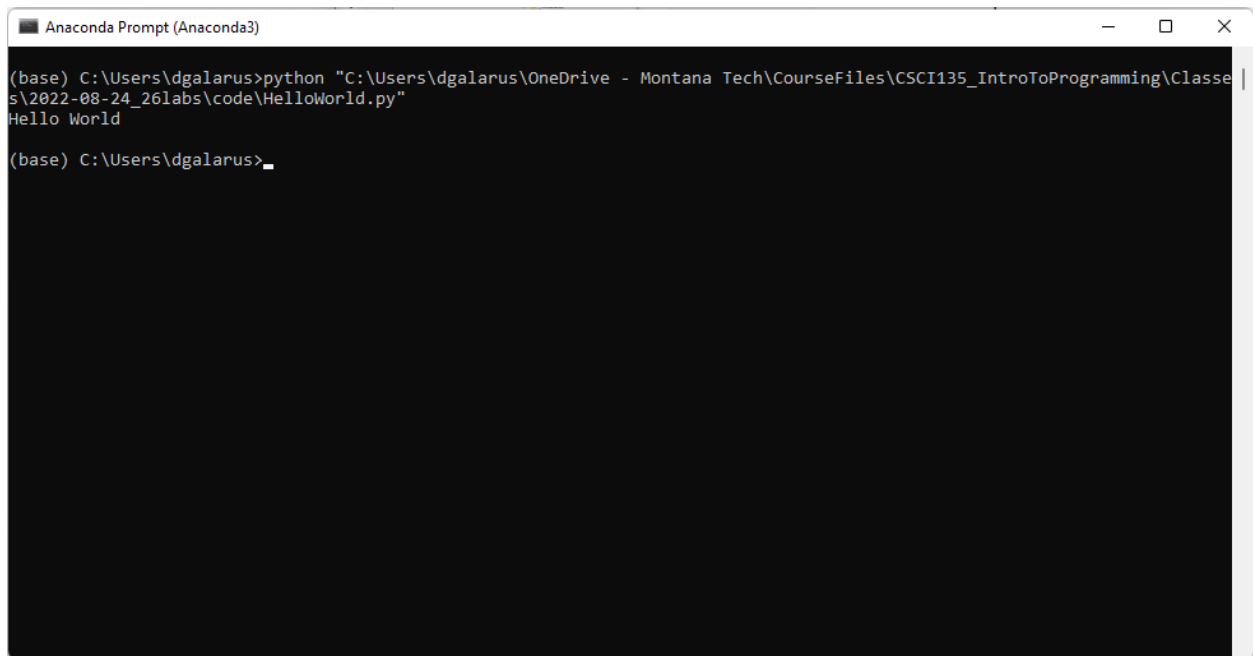
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Instead of navigating to the directory containing the Python script, an alternative is to include the full path to the script when executing it with the python command. In the following screenshots I copy a path to the file and then specify the full path, including the file name, when calling the python command.



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The script executes just like what was shown previously.

A screenshot of an Anaconda Prompt window. The title bar reads "Anaconda Prompt (Anaconda3)". The command prompt shows the following sequence of text: a prompt "(base) C:\Users\dgalarus>" followed by the command "python "C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code\HelloWorld.py\"", then the output "Hello World", and finally a new prompt "(base) C:\Users\dgalarus>".

```
(base) C:\Users\dgalarus>python "C:\Users\dgalarus\OneDrive - Montana Tech\CourseFiles\CSCI135_IntroToProgramming\Classes\2022-08-24_26labs\code\HelloWorld.py"
Hello World
(base) C:\Users\dgalarus>
```

Upload HelloWorld.py to Moodle as the first part of your submission for this assignment.

Part 2: Greetings from me (you)

Now you will apply what you learned in Part 1 to create a Python script that introduces yourself to whoever runs the script. Complete the following steps:

- Create a Python script named GreetingsFromMe.py.
- Include an appropriate header following the example we used for the Hello World script.
- Output / print your first and last name as the first line of output: “Hello, my name is First Last.”, substituting your first and last name.
- Output a blank line as the second line of output. You need to figure out how to do this.
- Output a third line telling us something interesting about yourself. Keep it clean, positive, interesting etc.

Upload GreetingsFromMe.py to Moodle as the second part of your submission for this assignment.

EXTENSIONS

I will try to add “extensions” to your assignments to give you further opportunity to explore, learn and practice. You will not submit the answers to these extensions as part of your assignment submission. You should work through and understand them to learn more and practice. View these an opportunity, not a burden

From the Liang book - NOTE: **We will not** use MyProgrammingLab in the book. You should be able to do exercises from Liang without MyProgrammingLab.:

1. *Answer the checkpoint and other inline questions in Chapter 1.*
2. *I may or may not cover use of Turtle graphics, as shown in Liang. You are welcome to try some of the Turtle graphics exercises and examples, though. The turtle module should be available with our Anaconda installs in the lab(s) and if you install Anaconda on your own computer.*
3. *Do programming exercises 1.1, 1.2 and 1.3, at the end of Chapter 1.*
4. *Try some other programming exercises at the end of Chapter 1.*

From the Things to Do section at the end of Chapter One of our optional Lubanovic book:

1. *If you don't already have Python 3 installed on your computer, do it now. Follow my separate instructions for installing the Anaconda Distribution.*
2. *Start the Python 3 interactive interpreter. Again, details are in Appendix B. It should print a few lines about itself and then a single line starting with >>>. That's your prompt to type Python commands.*
3. *Experiment with the interpreter. Use it like a calculator and type this: 8 * 9. Press the Enter key to see the result. Python should print 72.*
4. *Type the number 47 and press the Enter key. Did it print 47 for you on the next line?*
5. *Now, type print(47) and press Enter. Did that also print 47 for you on the next line?*

From me:

Create a Python script identical to archive.py shown in Example 1-4, pp 8-9, of the Lubanovic book. Note that the numbers at the start of each line in that book are not part of the code – don't type them in your file – these are line numbers used to reference each line in our book. I showed this code in class slides too. You can retype the contents in a new text file or download the file from our author's Github site:

<https://github.com/madscheme/introducing-python/blob/master/ch01/archive.py>

Test archive.py on various web addresses and dates to verify that it works.

Go through our author's explanation of each line in archive.py to better understand how the program works.

Add the following code to a new line in the script placed between lines 11 and 12: `print(text)`

See what gets printed for various sites and dates when you run the revised script. The line you added should show the JSON text response sent from archive.org in response to your request.

Try to interpret the JSON data that is shown in the output.